MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **PRE-BOARD MOCK EXAMINATION – 12 (LINEAR EQUATIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 14(iv).
2. **Sec A:** 4 MCQs (1M). **Sec B:** 2 AR (1M). **Sec C:** 3 VSAQ (2M). **Sec D:** 3 SA (3M). **Sec E:** 1 LA (5M). **Sec F:** 1 Case Study (4M).

SECTION A: Multiple Choice Questions (4 × 1 = 4 Marks)

Q1. The pair of linear equations $2x + 5y = k$ and $kx + 15y = 18$ has infinitely many solutions if: [1]
(A) $k = 3$ (B) $k = 6$ (C) $k = 9$ (D) no value of k exists

Q2. If $4x + 6y = 32$ and $4x - 2y = 8$, then the value of $8y$ is: [1]
(A) 24 (B) 16 (C) 32 (D) 8

Q3. The graphs of $x = 4$ and $y = -3$ intersect at: [1]
(A) $(-3, 4)$ (B) $(4, -3)$ (C) $(0, 0)$ (D) $(4, 0)$

Q4. The difference between two numbers is 26 and one number is three times the other. The numbers are: [1]
(A) 39 and 13 (B) 30 and 4 (C) 36 and 10 (D) 42 and 16

SECTION B: Assertion-Reason Questions (2 × 1 = 2 Marks)

Directions: (A) Both (A) and (R) are true, and (R) is correct explanation. (B) Both true, but (R) is not correct explanation. (C) (A) is true, (R) is false. (D) (A) is false, (R) is true.

Q5. **Assertion (A):** A two-digit number having tens digit x and units digit y is represented as $10x + y$.
Reason (R): When the digits are reversed, the new number formed is $10y + x$. [1]

Q6. **Assertion (A):** The system of linear equations $x + 2y - 4 = 0$ and $2x + 4y - 12 = 0$ has no solution.
Reason (R): Two parallel lines never intersect each other and hence have no common point of intersection. [1]

SECTION C: Very Short Answer Questions (3 × 2 = 6 Marks)

Q7. Determine whether the lines $3x + 2y = 5$ and $2x - 3y = 7$ are consistent or inconsistent. [2]

Q8. The sum of two numbers is 85. If their difference is 35, find the numbers. [2]

Q9. The perimeter of an isosceles triangle is 11 cm and its unequal base is 5 cm. Find the equal sides. [2]

SECTION D: Short Answer Questions (3 × 3 = 9 Marks)

Q10. Solve for x and y: $mx - ny = m^2 + n^2$ and $x + y = 2m$. [3]

Q11. The ratio of incomes of two persons is $9 : 7$ and the ratio of their expenditures is $4 : 3$. If each of them manages to save Rs. 2000 per month, find their monthly incomes. [3]

Q12. A two-digit number is 4 times the sum of its digits and twice the product of its digits. Find the number. [3]

SECTION E: Long Answer / HOTS Question (1 × 5 = 5 Marks)

Q13. A sailor can row a boat at speed x km/h in still water and the river flows at speed y km/h. The boat travels 28 km downstream in 2 hours and takes 4 hours to return the same distance upstream. Formulate a pair of linear equations in x and y, find the speed of the sailor in still water and the speed of the current, and find the total time taken to row 35 km in still water. [5]

SECTION F: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q14. Case Study: Railway Reservation Tariff & Concession Rules

Under Indian Railways passenger tariff rules, a child aged between 5 and 11 years is eligible for a half-fare ticket. However, the full reservation surcharge and superfast charge apply equally to both adult (full) and child (half) tickets.

A passenger booked 1 full adult first-class AC ticket from New Delhi to Varanasi and paid Rs. 2530 ($F + R = 2530$).

On another booking for the same journey, a family booked 1 full ticket and 1 half-fare ticket and paid a total sum of Rs. 3810 ($1.5F + 2R = 3810$).

Based on the above passenger fare structure, answer the following questions:

(i) Formulate the pair of linear equations in basic fare F and reservation charge R. [1]

(ii) Determine the full basic first-class AC fare (F). [1]

(iii) Determine the fixed reservation charge per ticket (R). [1]

(iv) Calculate the total ticket price for a family consisting of 2 adults and 2 children travelling on the same route. [1]

OR

Find the total amount saved by booking 1 half ticket instead of 1 full ticket. [1]

MATHS MANTHAN
TUITION CLASSES

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★